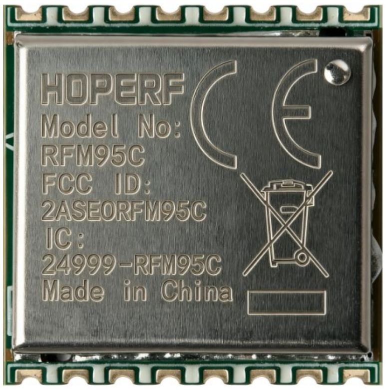
RFM95CW-868/915MHz LoRa Transceiver Module

General Description

RFM95CW is an ultra-low-power, high LoRa transceiver for various frequency of 137 MHz wireless applications. It is part of the SEMTECH RF product line, which includes complete transmitters, receivers and transceivers. The high integration of RFM95CW peripheral materials needed in the system design. The sensitivity up to - 138 dBm which can optimize the link performance of applications. In addition, RFM95CW also supports Duty-Cycle operation mode, channel interception, high power-on reset, noise output and other more functions, which makes the application design more flexible thus to achieve product differentiation design. The working voltage of RFM95CW is 1.8V~3.7V. When the sensitivity is reaching -138dBm, it only consumes 9.9 mA current. This ultra-low power mode can further reduce the power consumption of the module.



RFM95CW-868

RFM95CW-915

Features

- Frequency Range: 137~1020MHz
- Modulation: LoRa
- Data Rate: 0.018~37.5 kbps
- Sensitivity: -138 dBm, BW=125KHz, SF=12
- Voltage Range: 1.8~3.7 V
- Receiving Current: 12.5 mA @ BW=125KHz
- Supports Ultra Low Power Receiving Mode
- Sleeping Current
- 160 nA, Duty Cycle = OFF
- 600 nA, Duty Cycle = ON
- 4-wire SPI Interface
- Supports Full-automatic Independent Working Mode

Applications

- Automatic meter Reading
- Home Security & Building automation
- ISM Band Data Communication
- Industrial Monitoring & Controlling
- Security System
- Remote Control Application
- Intelligent Instrument
- Supply Chain & Logistics
- Intelligent Agriculture
- Smart City
- Retailing
- Asset Following
- Smart Lighting System
- Smart Parking
- Environmental Monitoring
- Health Monitoring

Product Pinout

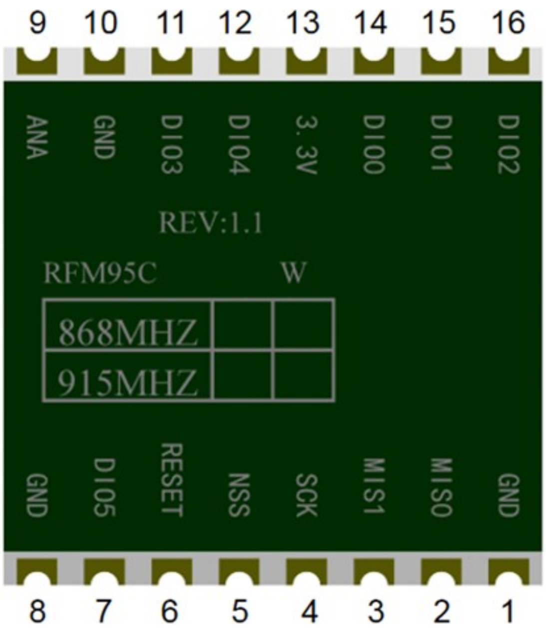


Figure 1. Bottom View of RFM95CW

Table1. Module Pin Definition of RFM95CW

Pin No.	Pin Name	Description
1	GND	Ground
2	MISO	SPI Data output
3	MOSI	SPI Data input
4	SCK	SPI Clock input
5	NSS	SPI Chip select input
6	RESET	Reset trigger input
7	DIO5	Digital I/O, software configured
8	GND	Ground
9	ANT	RF signal output/input
10	GND	Ground
11	DIO3	Digital I/O, software configured
12	DIO4	Digital I/O, software configured
13	3.3V	Supply voltage
14	DIO0	Digital I/O, software configured
15	DIO1	Digital I/O, software configured
16	DIO2	Digital I/O, software configured

Electrical parameters

Testing conditions: Power supply 3.3V, temperature 25℃

Table 2. Recommended Operating Conditions

Parameter	Symbol	Conditions	Minimum	Typical Value	Maximum	Unit
Supply Voltage	VDD		1.8	3.3	3.7	V
Operating Temperature	T		-40	85	℃	
Power Supply Voltage Slope				1		mV/us

Table 3. Absolute Maximum Rating

Parameter	Symbol	Conditions	Minimum	Maximum	Unit
Supply Voltage	VDD		-0.5	3.9	V
Interface Voltage	VIN		-0.3	3.3	V
Junction Temperature	TJ		-40	125	℃
Storage Temperature	TSTG		-50	150	℃
Soldering Temperature	TSDR	Last for at least 30s		255	℃
ESD Level[2]	HBM		-2	2	kV
Latch Current	@ 85 ℃		-100	100	mA

Table 4. Receiving Parameters

Parameter	Conditions	Minimum	Typical Value	Maximum	Unit
Frequency Tolerance	868 MHz 915 MHz	867.988 914.988	868 915	868.012 915.012	MHz
Transmitting Power	868MHz band 915MHz band	- -	18.3 18.3	- -	dBm
Power Reduction	16.3dBm Vbat=2.7V 15.3dBm Vbat=2.4V 12.3dBm Vbat=1.8V	- - -	2 3 6	- - -	dBm
Transmitting Current	868MHz 915MHz	- -	124 134	140 140	mA

Table 5. Receiving Parameters

Parameter	Conditions	Minimum	Typical Value	Maximum	Unit
Receiving Sensitivity (Lora) SF12, BW 125KHz CR4/5	868 MHz 915 MHz	- -	-138 -138	- -	dBm

Reference Design

Note that all schematics shown in this section are partial schematics, maybe not listing ALL required components.

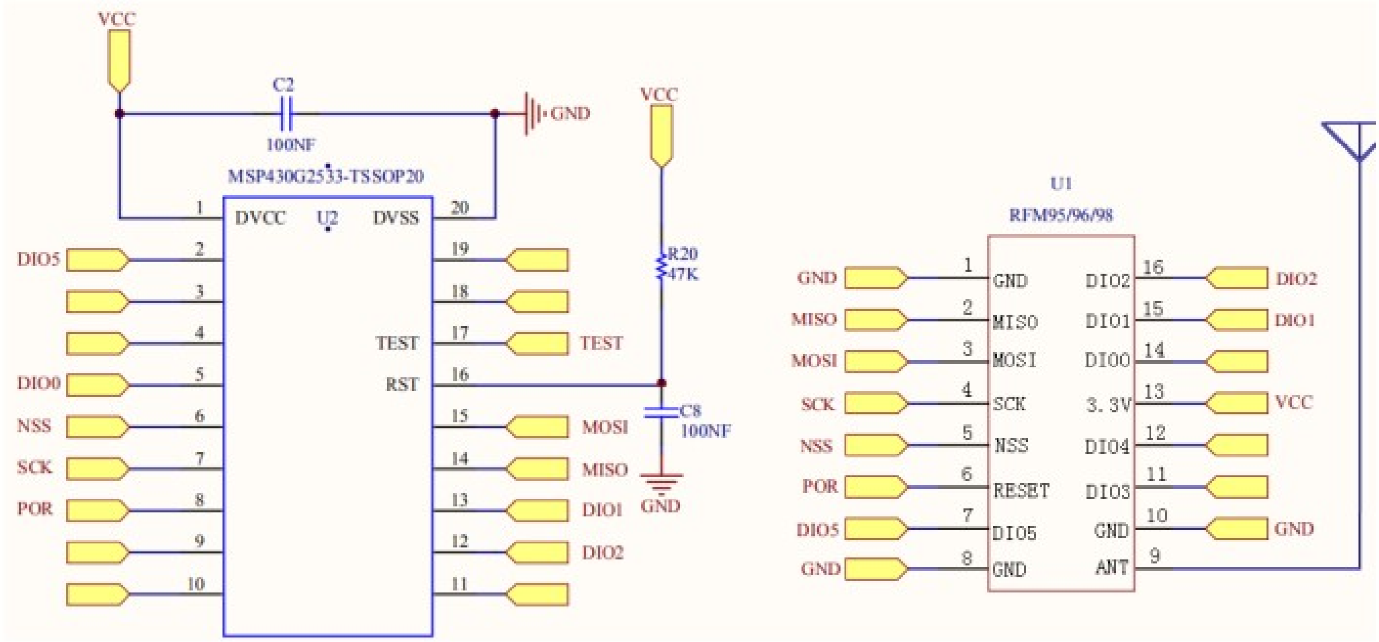


Figure 2. Reference Schematic

Module Size

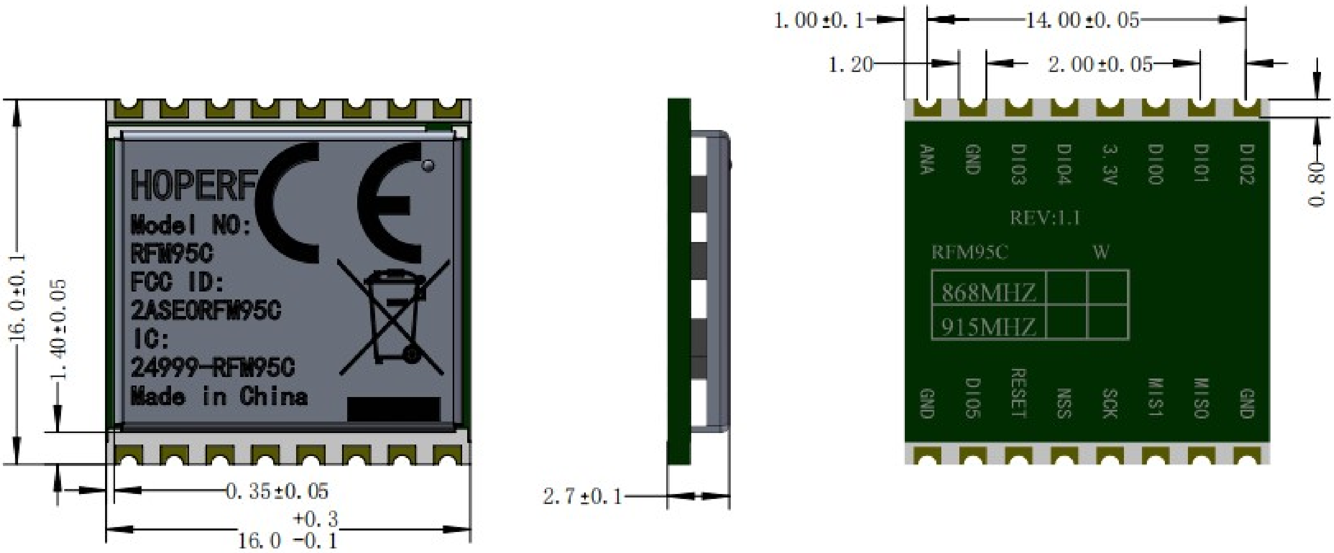


Figure 3. RFM95CW-868S2 Module Size (Unit: mm)

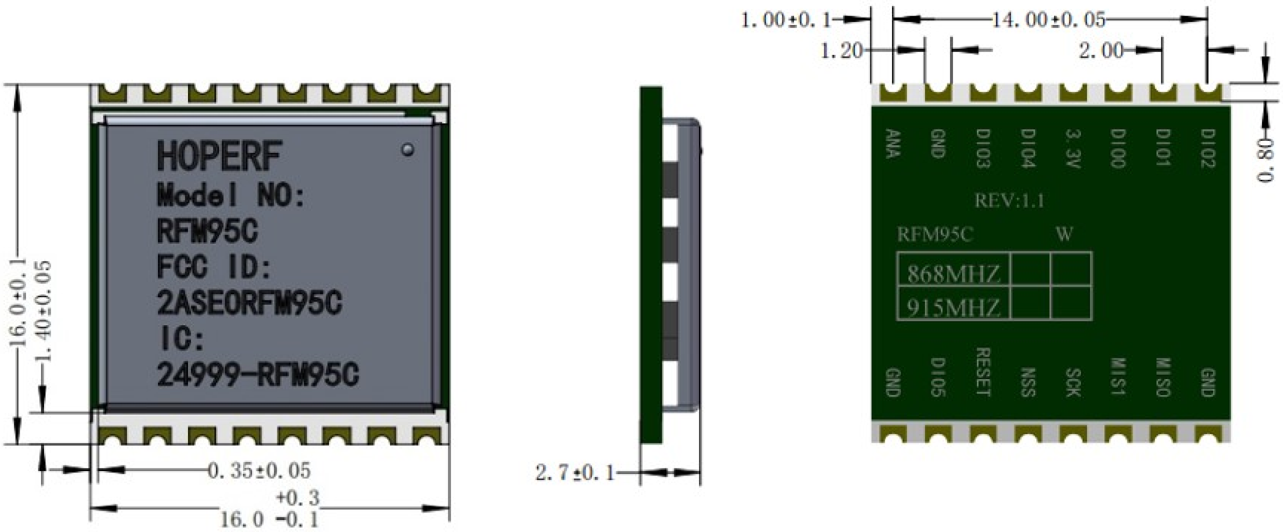
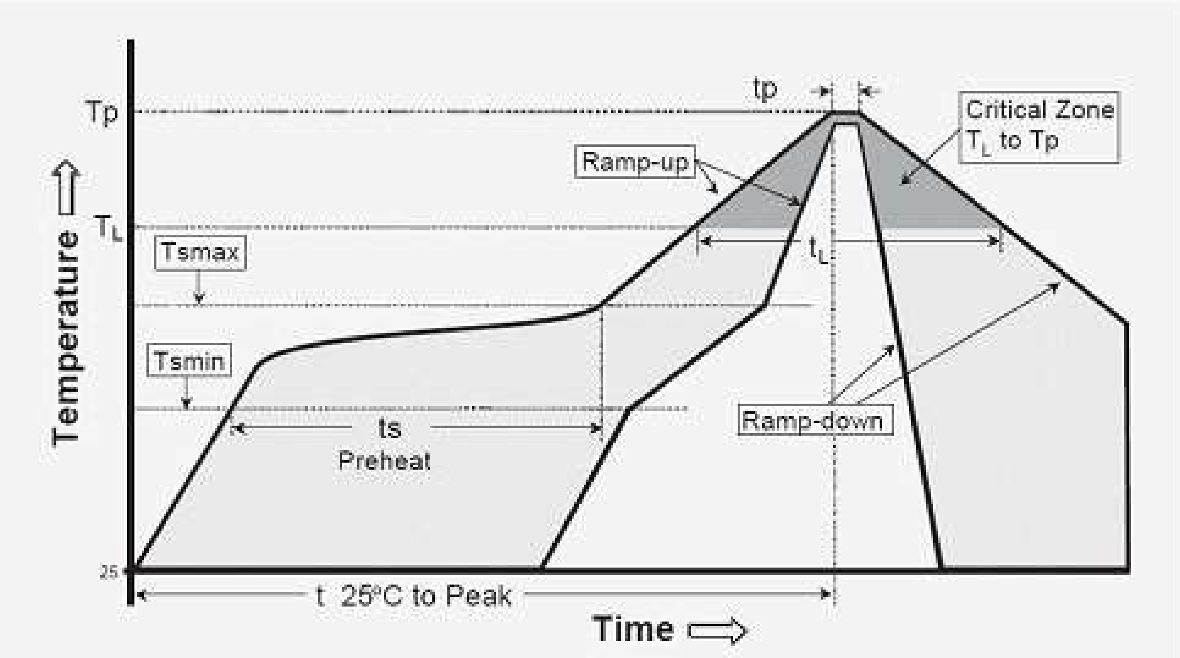


Figure 4. RFM95CW-915S2 Module Size (Unit: mm)

Reflow soldering temperature curve

It is recommended to comply with the related IPC standards and depend on the actual curve of the solder paste to setup the reflow soldering temperature curve.

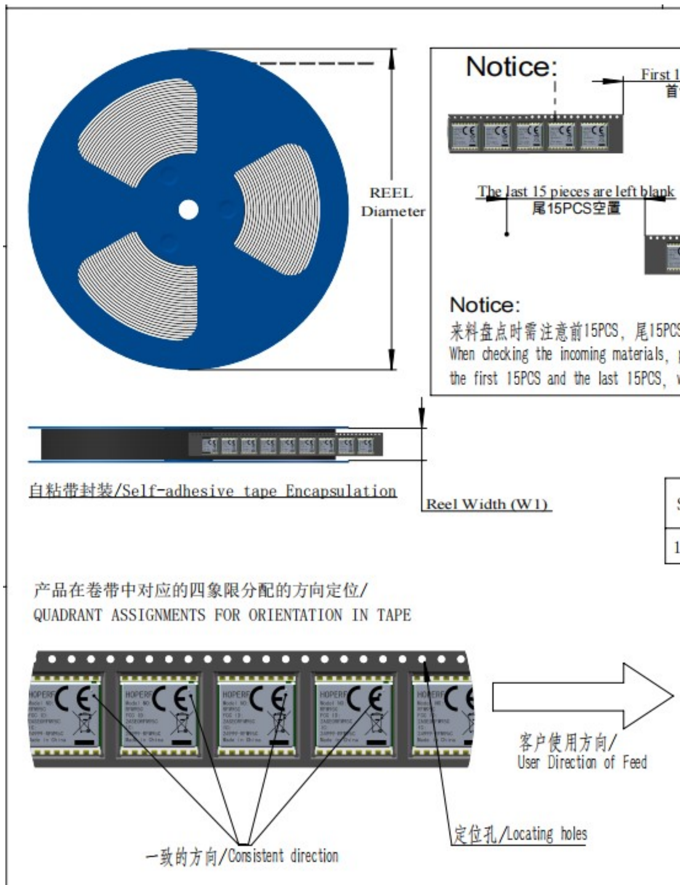


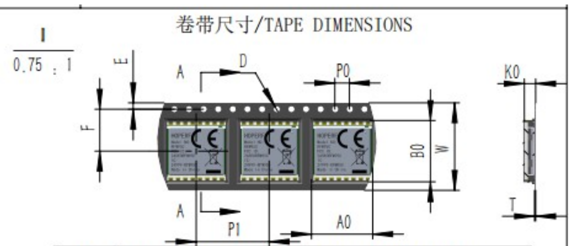
Pre-heating temperature - Minimum temperature (Tsmin) - Maximum temperature (Tsmax) - Warm-up time (ts)	 150 ℃ 200 ℃ 60 ~ 180 sec
Temperature ramping rate for preheating to the peak temperature（TL-TP）	3 ℃/sec (maximum)
Soldering time - Soldering temperature - Time to reach above the melting point	60 ~ 90 sec 217 ℃ 60 ~ 150 sec
Peak temperature	240 +/- 5 ℃
Soldering time for peak temperature +/- ℃	10~30 sec
Cooling rate	6 ℃/sec (maximum)
Full soldering time	8 min (maximum)

Notes

- The wettability of lead-free solder materials is worse than that of tin-lead eutectic materials, and the melting zone temperature is higher. To achieve better soldering results and protect the solder joints from oxidation, use nitrogen protection during reflow soldering.
- For double-sided products, avoid reflow soldering with the module facing downward (that is, the module should be soldered during the second reflow soldering) to prevent the components on the module from falling off or shifting.
- The module needs to be fully tinned, and it is recommended to use a stepped steel mesh with a thickness of 0.15MM or above.
- When the module must be re-soldered, we recommend manual soldering to avoid possible damage to the module caused by multiple reflow soldering processes.
- Customers should develop welding processes that comply with the above tips based on actual production conditions.

Packaging Information





A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers
E	Distance from locating hole to edge
F	Distance from locating hole to product center
T	Thickness of the roll
P0	Center Distance of locating hole
D	Hole Diameter of locating hole

Reel width W1 (±0.3) ; W0 (±0.3) ; ther Dimension Tolerance(±0.1)

SPQ	Reel diameter (mm)	Reel width (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	E (mm)	F (mm)	T (mm)	P0 (mm)	D (mm)	Pin1 Quad
1000	330	25.1	16.4	16.4	3.1	20	24	1.75	11.5	0.3	4	1.5	

Technical Requirements:

1. The cumulative tolerance of every 10 cavity spacing is ±0.2mm;
2. Surface friction voltage <10V,
   10*10ˆ4 ohms < surface resistance <1*10ˆ9 ohms;
   The product is encapsulated with selfadhesive tape;
3. All sizes meet the international standard EIA-481-3;
4. The number of products packaged per 13-inch is 1030 PCS;
5. Material color: Black;
6. The bending of the tape shall not exceed 1mm per 100mm;
7. The width tolerance of the reel and the tape W1, W0 is ±0.3;
   A0, B0 measured with the upper surface of the Mold cavity as the datum, with a tolerance of ±0.1 mm;
   K0 value is measured as the value from the bottom of the mold cavity to the upper surface of the carrier.

Ordering Information

Model No.	Working Frequency	Certifications
RFM95CW-868S2	868 MHz	CE(EU)
RFM95CW-915S2	915 MHz	FCC(USA)、ISED(Canada)

Revision History

Version	Date	Description
V1.0	2024.3	Initial release
V1.1	2024.10.31	Update module picture
V1.2	2025.05.26	Update the PIN NAME of the antenna
V1.3	2025.07.29	Update the relevant information of the 868MHz module
V1.4	2026.03.12	Update the pin numbers and the pin descriptions; Update reflow soldering temperature curve，reference designs,packaging information and certification information. Update the module photos.

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